PRACTICE SET

FangMath

Getting Ready for Similarity

Similar rectangles and similar triangles

Name: _____ Date: _____

Part A Similar Rectangles

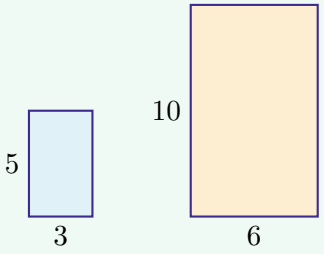
Key Idea

Two shapes are **similar** when one is an enlargement (or shrink) of the other: **same shape, different size**. For rectangles, **long side** $\leftrightarrow$ **long side** and **short side** $\leftrightarrow$ **short side**. The rectangles are similar when both ratios are the same number:

$$\frac{\text{long side of 2nd}}{\text{long side of 1st}} = \frac{\text{short side of 2nd}}{\text{short side of 1st}} = \text{scale factor}$$

Watch Me

Example 1 — Are they similar?



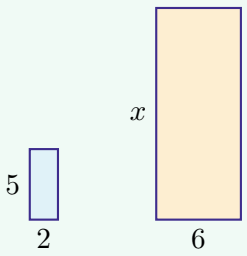
Long sides: $\frac{10}{5} = 2$ Short sides: $\frac{6}{3} = 2$

Both ratios are 2, so the rectangles **are similar**.

Scale factor $= \mathbf{2}$.

Watch Me

Example 2 — Find the missing side. The rectangles are similar.



Method 1 — Scale factor

Use the pair you know: $\frac{6}{2} = 3$, so the scale factor is 3.

Multiply: $x = 5 \times 3 = \mathbf{15}$.

Method 2 — Proportion

Write $\frac{\text{2nd}}{\text{1st}}$ for both pairs of matching sides:

$$\frac{6}{2} = \frac{x}{5} \Rightarrow 2x = 6 \times 5 = 30 \Rightarrow x = \mathbf{15}$$

Common Mistake

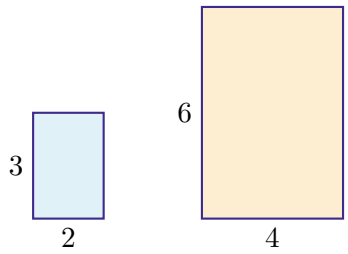
2×3 and 4×5 are **not** similar. Adding 2 to every side changes the shape. Similar shapes come from **multiplying** every side by the same number.

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EASY **Rectangles**

Problems 1–5: Are the rectangles similar? If yes, give the scale factor (1st → 2nd).

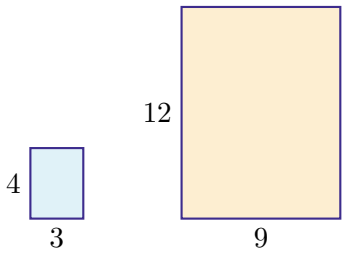
1.



Long sides: ___ / ___ Short sides: ___ / ___

Similar? Yes / No Scale factor: _____

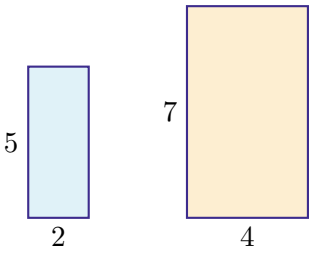
2.



Long sides: ___ / ___ Short sides: ___ / ___

Similar? Yes / No Scale factor: _____

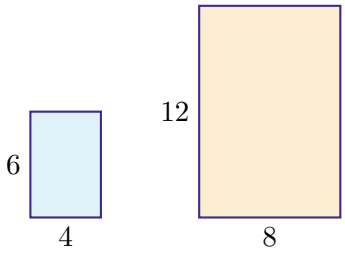
3.



Long sides: ___ / ___ Short sides: ___ / ___

Similar? Yes / No Scale factor: _____

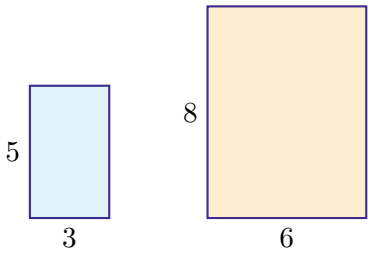
4.



Long sides: ___ / ___ Short sides: ___ / ___

Similar? Yes / No Scale factor: _____

5.



Long sides: ___ / ___ Short sides: ___ / ___

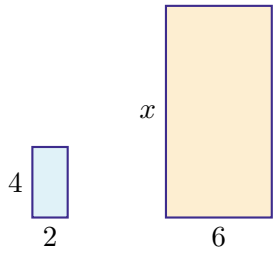
Similar? Yes / No Scale factor: _____

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EASY Rectangles — Find the Missing Side

The rectangles are similar. Find x two ways: with the scale factor and with a proportion.

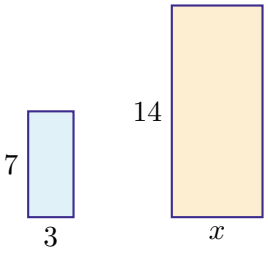
6.



Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{____}{____} = \frac{____}{____}$ $x =$ _____

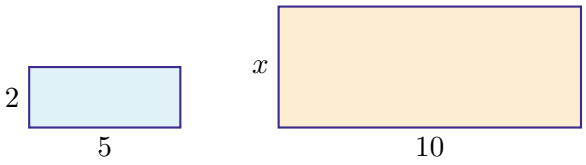
7.



Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{____}{____} = \frac{____}{____}$ $x =$ _____

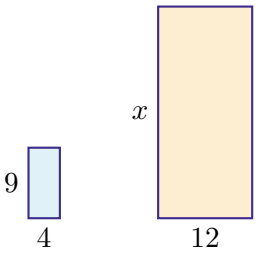
8.



Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{____}{____} = \frac{____}{____}$ $x =$ _____

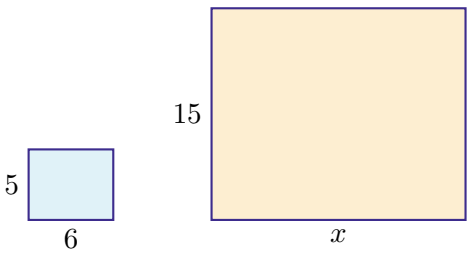
9.



Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{____}{____} = \frac{____}{____}$ $x =$ _____

10.



Method 1 — Scale factor: _____ $x =$ _____

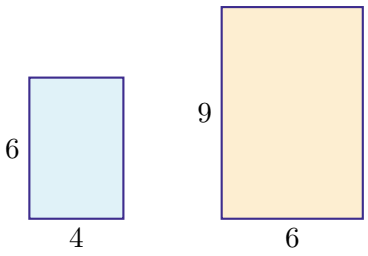
Method 2 — Proportion: $\frac{____}{____} = \frac{____}{____}$ $x =$ _____

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MEDIUM **Rectangles**

Problems 1–5: Are the rectangles similar? Watch out — a rectangle may be turned sideways, and the scale factor may be a fraction.

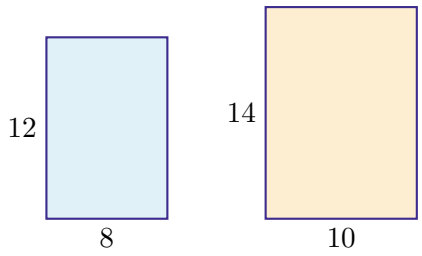
1.



Long sides: ___/___ Short sides: ___/___

Similar? Yes / No Scale factor: _____

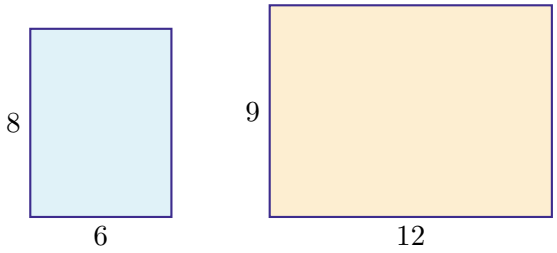
2.



Long sides: ___/___ Short sides: ___/___

Similar? Yes / No Scale factor: _____

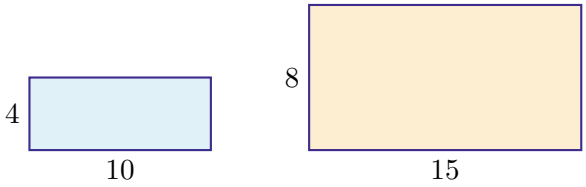
3.



Long sides: ___/___ Short sides: ___/___

Similar? Yes / No Scale factor: _____

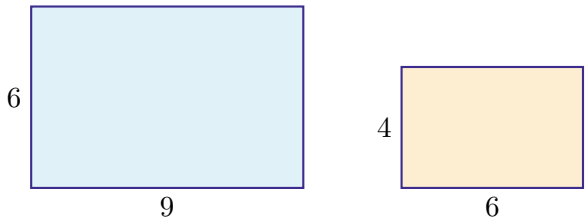
4.



Long sides: ___/___ Short sides: ___/___

Similar? Yes / No Scale factor: _____

5.



Long sides: ___/___ Short sides: ___/___

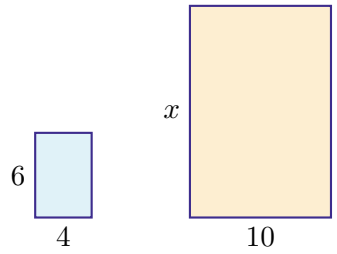
Similar? Yes / No Scale factor: _____

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MEDIUM **Rectangles — Find the Missing Side**

The rectangles are similar. Find x two ways. The scale factor may be a fraction, and x may be on the 1st rectangle.

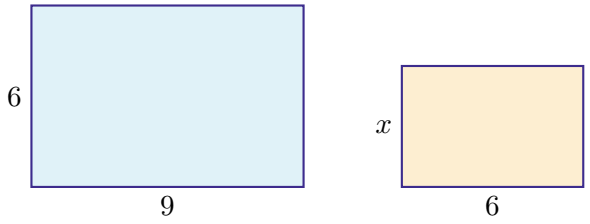
6.



Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{____}{____} = \frac{____}{____}$ $x =$ _____

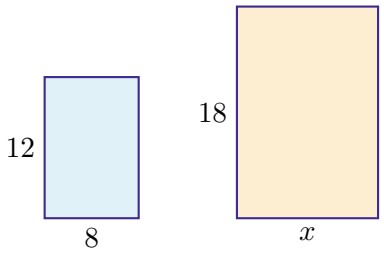
7.



Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{____}{____} = \frac{____}{____}$ $x =$ _____

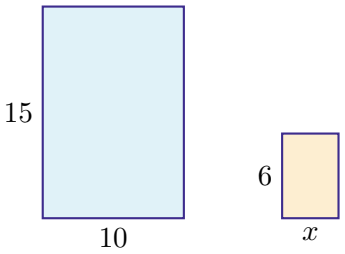
8.



Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{____}{____} = \frac{____}{____}$ $x =$ _____

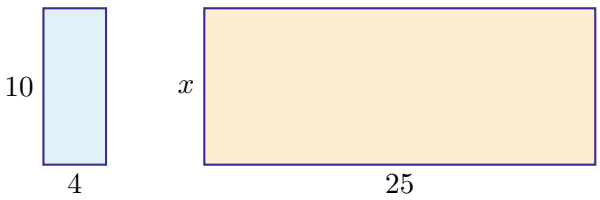
9.



Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{____}{____} = \frac{____}{____}$ $x =$ _____

10.



Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{____}{____} = \frac{____}{____}$ $x =$ _____

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Part B Similar Triangles

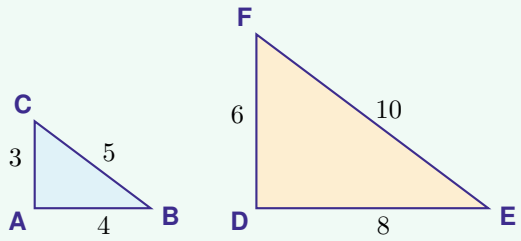
Key Idea

Match the sides by size: **shortest** $\leftrightarrow$ **shortest, middle** $\leftrightarrow$ **middle, longest** $\leftrightarrow$ **longest**. The triangles are similar when **all three** ratios are the same number (the scale factor).

In a **similarity statement** $\triangle ABC \sim \triangle DEF$, the **order** of the letters shows which vertices match: $A \leftrightarrow D$, $B \leftrightarrow E$, $C \leftrightarrow F$.

Watch Me

Example 1 — Are they similar?

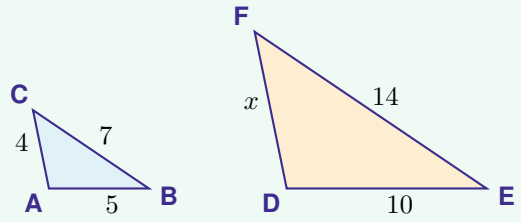


Shortest: $\frac{6}{3} = 2$ Middle: $\frac{8}{4} = 2$ Longest: $\frac{10}{5} = 2$

All three ratios are 2, so the triangles **are similar**: $\triangle\mathbf{ABC} \sim \triangle\mathbf{DEF}$, scale factor $\mathbf{2}$.

Watch Me

Example 2 — Find the missing side. $\triangle ABC \sim \triangle DEF$.



Method 1 — Scale factor

$AB \leftrightarrow DE$: $\frac{10}{5} = 2$.

$x = FD$ matches $CA = 4$: $x = 4 \times 2 = \mathbf{8}$.

Method 2 — Proportion

$$\frac{DE}{AB} = \frac{FD}{CA}$$

$$\frac{10}{5} = \frac{x}{4} \Rightarrow 5x = 40 \Rightarrow x = \mathbf{8}$$

Common Mistake

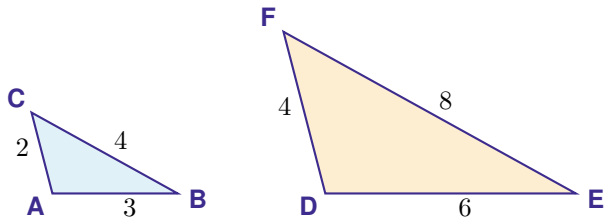
Checking only **two** pairs of sides is not enough. Sides $4, 6, 3$ and $8, 10, 6$ have two ratios equal to 2, but $\frac{10}{6} \neq 2$ — so they are **not** similar. Check all three!

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EASY **Triangles**

Problems 1–5: Are the triangles similar? If yes, write a similarity statement.

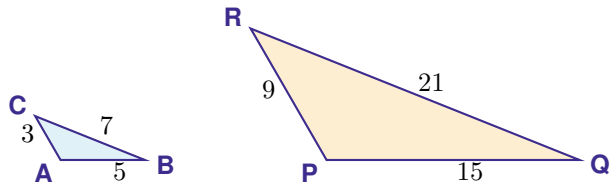
1.



Shortest: ____ Middle: ____ Longest: ____

Similar? Yes / No △_____ ~ △_____

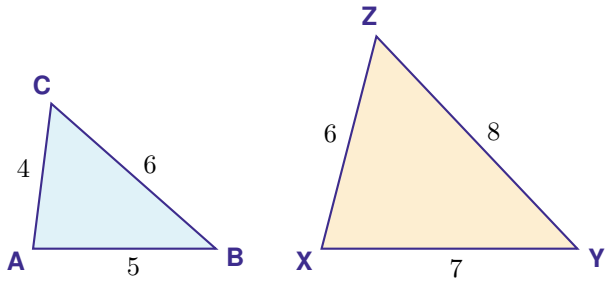
2.



Shortest: ____ Middle: ____ Longest: ____

Similar? Yes / No △_____ ~ △_____

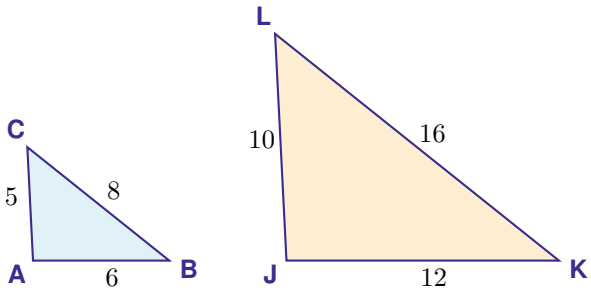
3.



Shortest: ____ Middle: ____ Longest: ____

Similar? Yes / No △_____ ~ △_____

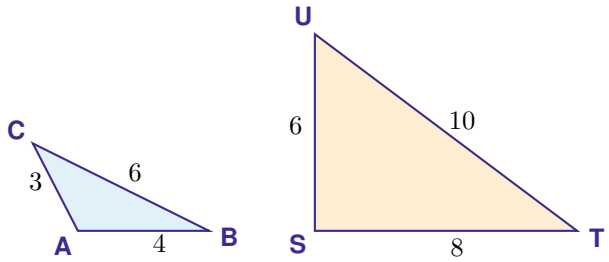
4.



Shortest: ____ Middle: ____ Longest: ____

Similar? Yes / No △_____ ~ △_____

5.



Shortest: ____ Middle: ____ Longest: ____

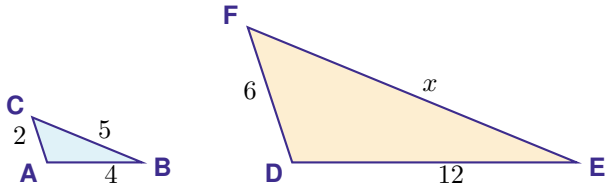
Similar? Yes / No △_____ ~ △_____

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EASY **Triangles — Find the Missing Side**

The triangles are similar. Find x two ways: with the scale factor and with a proportion.

6.

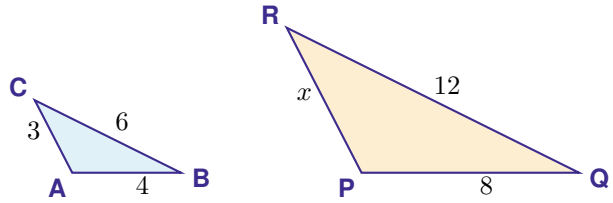


$\triangle ABC \sim \triangle DEF$

Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{____}{____} = \frac{____}{____}$ $x =$ _____

7.

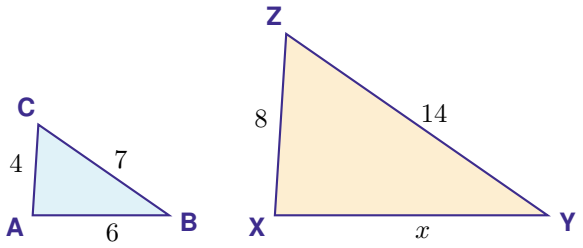


$\triangle ABC \sim \triangle PQR$

Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{____}{____} = \frac{____}{____}$ $x =$ _____

8.

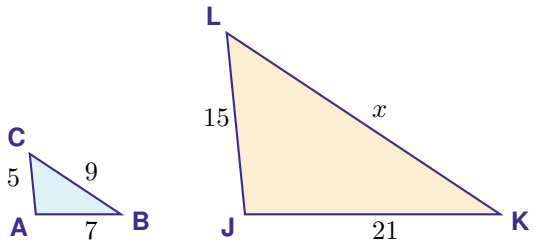


$\triangle ABC \sim \triangle XYZ$

Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{____}{____} = \frac{____}{____}$ $x =$ _____

9.

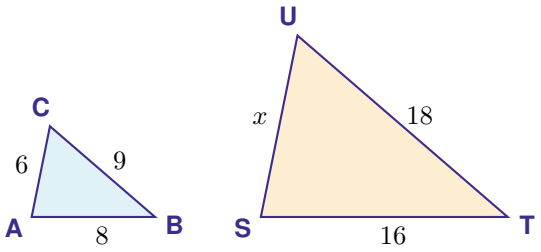


$\triangle ABC \sim \triangle JKL$

Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{____}{____} = \frac{____}{____}$ $x =$ _____

10.



$\triangle ABC \sim \triangle STU$

Method 1 — Scale factor: _____ $x =$ _____

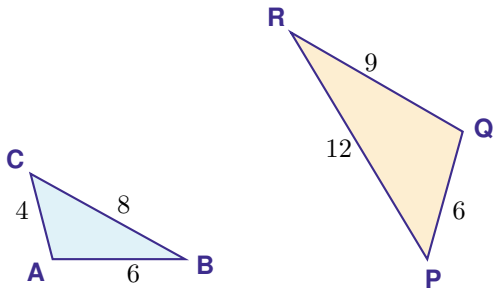
Method 2 — Proportion: $\frac{____}{____} = \frac{____}{____}$ $x =$ _____

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MEDIUM **Triangles**

Problems 1–5: Are the triangles similar? A triangle may be turned or flipped, so match sides by **size**, not by position.

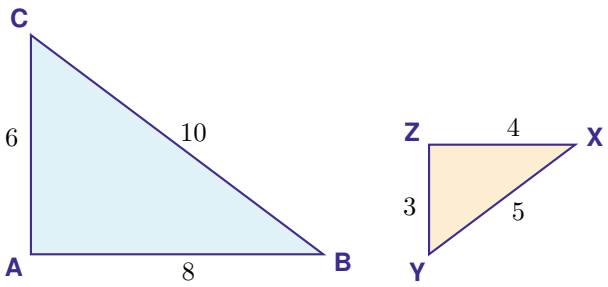
1.



Shortest: ____ Middle: ____ Longest: ____

Similar? Yes / No △_____ ~ △_____

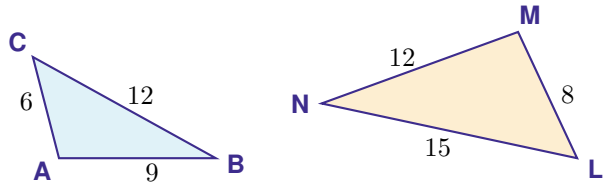
2.



Shortest: ____ Middle: ____ Longest: ____

Similar? Yes / No △_____ ~ △_____

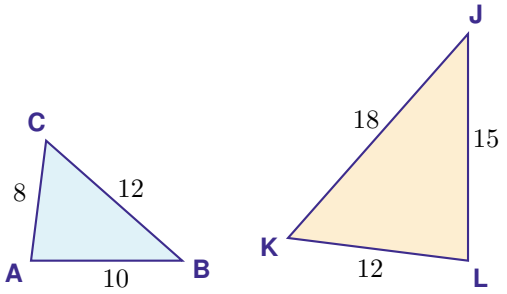
3.



Shortest: ____ Middle: ____ Longest: ____

Similar? Yes / No △_____ ~ △_____

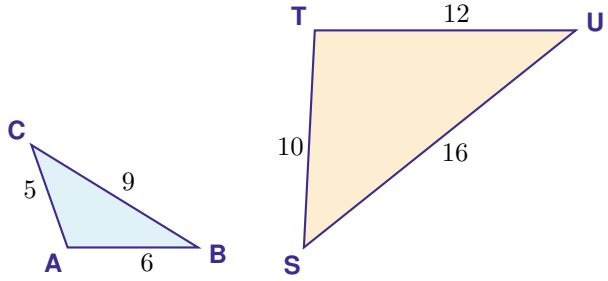
4.



Shortest: ____ Middle: ____ Longest: ____

Similar? Yes / No △_____ ~ △_____

5.



Shortest: ____ Middle: ____ Longest: ____

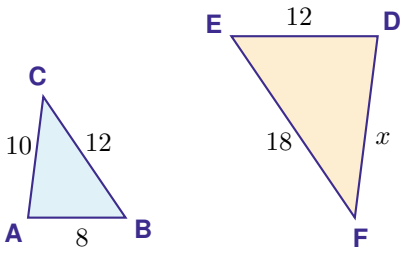
Similar? Yes / No △_____ ~ △_____

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MEDIUM **Triangles — Find the Missing Side**

Use the similarity statement to match sides (the triangles may be turned or flipped), then find x two ways.

6.

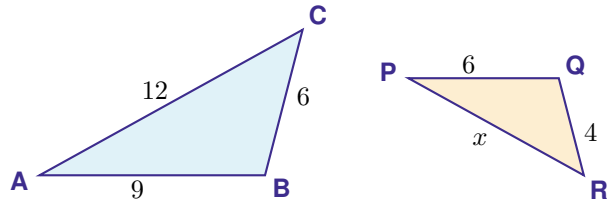


$\triangle ABC \sim \triangle DEF$

Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{\quad}{\quad} = \frac{\quad}{\quad}$ $x =$ _____

7.

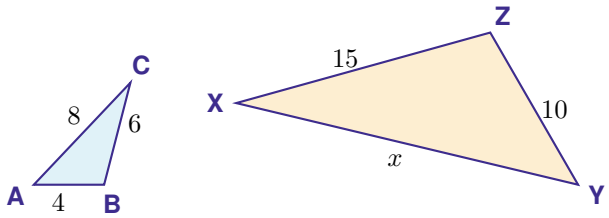


$\triangle ABC \sim \triangle PQR$

Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{\quad}{\quad} = \frac{\quad}{\quad}$ $x =$ _____

8.

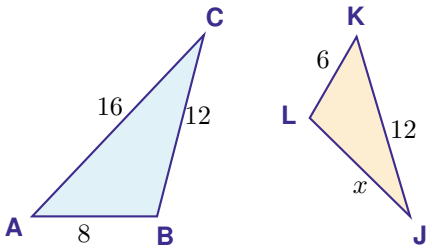


$\triangle ABC \sim \triangle YZX$

Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{\quad}{\quad} = \frac{\quad}{\quad}$ $x =$ _____

9.

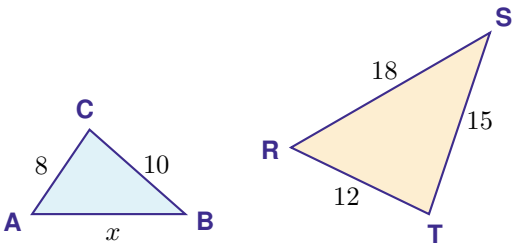


$\triangle ABC \sim \triangle KLJ$

Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{\quad}{\quad} = \frac{\quad}{\quad}$ $x =$ _____

10.



$\triangle ABC \sim \triangle RST$

Method 1 — Scale factor: _____ $x =$ _____

Method 2 — Proportion: $\frac{\quad}{\quad} = \frac{\quad}{\quad}$ $x =$ _____

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